

App Dev Project Report

1. Student Details

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About Me: I am a student at IIT Madras BS Degree program with a deep interest in web application development and software engineering. I enjoy building meaningful applications that solve real-world problems.

2. Project Details

Project Title: Placement Portal Application (PPA)

Problem Statement: To design and build a multi-role web-based application that allows Admin (Institute), Companies, and Students to efficiently manage campus recruitment activities, including company approvals, placement drives, student applications, and background jobs.

Approach: The app was built using Flask as the backend framework with a modular Blueprint structure and Vue.js for the frontend SPA. It allows role-based access control, drive management, automatic eligibility filtering, and background jobs via Celery/Redis for notifications and CSV exports.

3. AI/LLM Declaration

Summary View (Category-wise Totals)

Category	MAD-2 Total %	Notes
Backend (Flask + DB)	32%	Slightly reduced in MAD-2 due to API-only model
Frontend (HTML/JS)	38%	Shift from Jinja to Vue-based architecture
Styling/UI	9%	No significant difference
Auth + CRUD Logic	14%	Logic moves from backend templates to APIs
Extras (Charts/API/etc)	7%	Still optional, depending on student projects

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Vue.js 3	Frontend framework for reactive SPA UI
Bootstrap 5	Frontend styling and responsive design
Celery / Redis	Asynchronous background task processing
Flask-JWT-Extended	User authentication and access tokens
Flask-Caching	Performance optimization via response caching
SQLite	Lightweight local database for storing application data

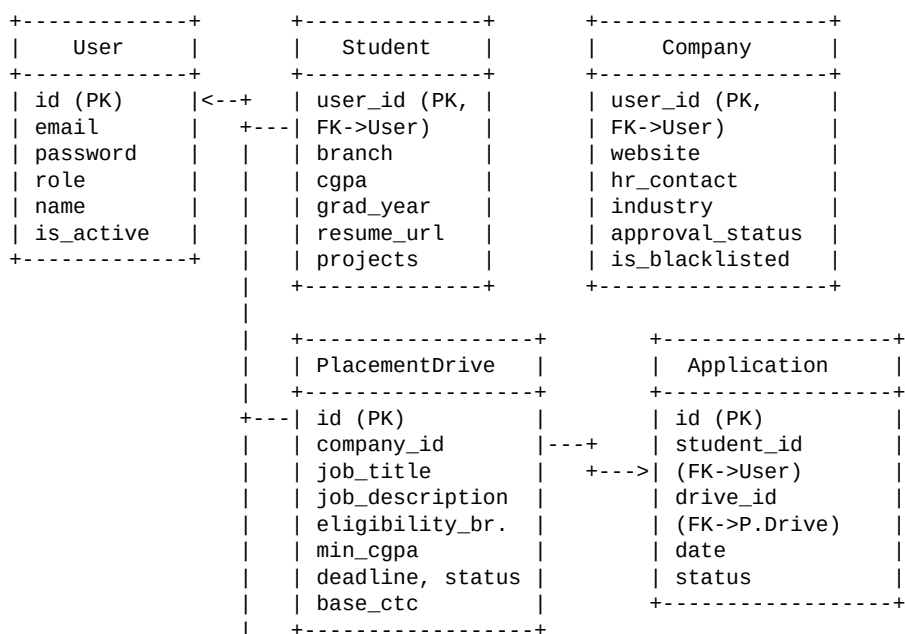
5. Database Schema / ER Diagram

Tables:

1. User - stores authentication and global details (id, email, password, role, is_active)
2. Student - stores student academic details (user_id, branch, cgpa, grad_year)
3. Company - stores employer profiles (user_id, website, hr_contact, industry)
4. PlacementDrive - stores job listings (id, company_id, job_title, status)
5. Application - logs student job applications (id, student_id, drive_id, status)
6. AuditLog - tracks admin actions (id, admin_id, action, timestamp)

Relationships:

- One-to-One -> User -> Student / Company
- One-to-Many -> Company (User) -> PlacementDrive
- One-to-Many -> Student (User) -> Application
- One-to-Many -> PlacementDrive -> Application



6. API Resource Endpoints

Endpoint	Method	Description
/api/auth/login	POST	Authenticate user and generate JWT session
/api/student/apply	POST	Add new job application entry for student
/api/company/drives	GET	Fetch placement drives posted by company
/api/admin/stats	GET	Return system-wide statistics for admin
/api/student/export	POST	Async task to export student logs as CSV

Note: Complete endpoint documentation handled dynamically within the source code.

7. Architecture and Features (optional)

Architecture Overview:

- app.py - main Flask application entry point with Celery configuration

- /models - database models using SQLAlchemy and extension hubs
- /routes - Flask Blueprints for auth, admin, company, and student endpoints
- /frontend - Vue.js components mounted to a single index.html

Implemented Features:

- Multi-role user registration and login (Admin, Company, Student)
- Placement drive creation and student eligibility checking
- Admin dashboard for approving users and overrides
- Advanced routing with Vue Router for client-side SPA
- Responsive UI built with Bootstrap 5
- Secure password hashing and JWT token management

Additional Features:

- Export user logs as CSV asynchronously via Celery
- Redis-backed caching for frequently accessed admin metrics
- Daily scheduled Celery tasks for deadline notifications

8. Video Presentation

Drive Link: <https://drive.google.com/file/d/1oh57F1wQCQ8LoG2h3RMQIFMyiUNSCoHQ/view?usp=sharing>